Ecological succession and Popper's historicism

Two questions: Can you find texts analyzing the concepts "ecological succession" and "climax community" from the point of view of Karl Popper's Historicism theory? Is any theory on this discussing philosophical theories of causality?

Searched the web

This is a rich area. Let me synthesize what I found.

First question: texts applying Popper's historicism critique to ecological succession / climax community

The short answer is: no text I can find does this *directly and explicitly* — i.e., nobody has published a paper titled "Clements and Popper's Historicism." But the connection is real and well-documented *obliquely*, and one paper comes very close.

The most relevant text is Christopher Eliot's **"Method and Metaphysics in Clements's and Gleason's Ecological Explanations"** (2007), published in *Studies in History and Philosophy of Biological and Biomedical Sciences* 38(1): 85–109. This paper directly invokes Popper in connection with Clements's climax theory: one of Gleason's criticisms of Clements was that his account adopted excessively many new terms, which may represent insulation against falsifiability. The paper explicitly notes that Popper famously worried about theories that neutralize exceptions through merely *ad hoc* amendments — and the analogy to Freudian theory is drawn. Eliot's deeper argument, however, is revisionist: examining Clements's approach to explanation reveals him not to be expressing a law at all, but instead developing an explanatory structure without laws, capable of progressively integrating causal complexity. Clements and Gleason largely agree on the *causes* of vegetation; their differences lie in methodology, not metaphysics. ScienceDirect ScienceDirect

This is important for the Popper connection. Popper's critique of historicism is precisely that historicists mistake a *trend* (a singular historical trajectory) for a *law* (a universal, falsifiable regularity). Popper writes that the evolution of life on earth is a unique historical process which "proceeds in accordance with all kinds of causal laws, for example, the laws of mechanics, of chemistry, of heredity and segregation, of natural selection, etc. — but its *description* is not a law, but only a single historical statement." The classical Clementsian climax doctrine — describing successional development in analogy to the ontogenetic development of individual organisms, culminating in a determined endpoint — maps almost perfectly onto what Popper calls historicist thinking: the projection of a singular developmental trajectory as if it were a law of nature. FS Blog Wikipedia

A second very relevant paper is **Annette Voigt's "How political philosophies can help to discuss and differentiate theories in community ecology"** (2021), in *History and Philosophy of the Life Sciences*. This paper uses structural analogies to competing political philosophies — liberalism, democratism, and conservatism — as a heuristic tool to differentiate between ecological theories, arguing that these two different areas of knowledge share in common that their objects are "societies" composed of individuals, and that contradictory and competing theories about these supra-individual units exist. While Voigt does not invoke Popper's Historicism directly, she is explicitly working in the tradition of using political-philosophical

frameworks to analyze ecological theory — precisely the framework in which Popper's critique of historicism is most naturally applicable. nih

A third relevant text is Philippe Huneman's **"The Chicago school of ecology's evolutionary superorganism and the Clements-Wright connection"** (2025, *History and Philosophy of the Life Sciences*). This paper investigates the fate of "organicism" — the idea that ecosystems or communities are, or are like, organisms — in major attempts at theoretical synthesis of ecology in the first part of the twentieth century, tracing how Clements's physiological and developmental concepts were subsequently reworked. The organicist model of the community-as-organism with a developmental "climax" endpoint is exactly the structure Popper would identify as quasi-historicist: a teleologically ordered, directional sequence culminating in a predetermined stable state. nih

Finally there is a paper arguing the opposite case — that Popperianism is not the right framework for ecology at all: **"Popperian ecology is a delusion"** (in *PLoS ONE*), which argues that Popper's falsificationism has been the most popular philosophy of science among ecologists, to the point where critics accuse its supporters of "Popperphilia," but that it is fundamentally an unsuitable philosophy of science for ecology. This is a useful critical counterpoint. nih

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Second question: is there work on theories of causality in this context?

Yes, this is explicitly addressed in the philosophy-of-ecology literature. The Oxford Bibliographies entry on Ecological Philosophy identifies "inference about causes, integration versus unification, function and teleology, identity and individuation of units, reduction and emergence, and historical science" as central conceptual issues linking ecology and philosophy of biology. Oxford Bibliographies

The Eliot paper again is most directly relevant here. Its central argument is that Clements and Gleason largely agree on what *causes* vegetation, but that "causal understanding here underdetermines representation" — meaning that both ecologists work from a similar causal background but differ in how they integrate causes into general theory. This is a philosophically sophisticated point about the relationship between causal knowledge and theory structure, and it connects directly to debates about mechanistic vs. covering-law models of explanation in the philosophy of science. ScienceDirect

The deeper issue is the distinction between **efficient causality** and **teleological/final causality** in Clements's framework. Clements wrote that "the explanation of the universal occurrence of a climax in succession lies in the fact that the succession is reproduction" — the reproductive process cannot fail to terminate in the adult form. This is frankly a teleological causal claim: the endpoint (climax/adult form) is invoked to explain the process. Popper's critique of historicism is closely related to

this: historicists, he argued, import teleological assumptions disguised as causal laws, confusing the description of a unique process with a universal predictive law. ScienceDirect

The directionality debate is addressed in recent ecological theory as well. A wide range of spatio-temporal patterns in ecological communities exhibit some degree of directionality and predictability, studied under the name of succession patterns. Ecological succession was originally pictured as a predictable sequence of community compositions, akin to an organism's developmental stages — but this simple picture was rapidly shown to present two major discrepancies with natural patterns: first, communities rarely display only a few discrete well-separated states; and second, changes in community composition are not necessarily fully deterministic or sequential, with some communities exhibiting stochastic and chaotic changes in dominance. biorxiv

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Summary and recommendations

The intersection you're asking about has not been treated in a single dedicated monograph or article, but the pieces are there. The most productive reading path would be:

1. **Eliot 2007** — the paper that comes closest, with an explicit Popper reference and deep engagement with causality in succession theory.
2. **Voigt 2021** — political philosophy applied to community ecology theory, in *History and Philosophy of the Life Sciences*.
3. **Huneman 2025** — the organicism genealogy, which contextualizes the teleological architecture of Clementsian thought.
4. **Popper, *The Poverty of Historicism*** (1957), especially the passages distinguishing causal laws from unique historical trends — read against Clements, the analogy is quite striking.
5. **"Popperian ecology is a delusion"** — as a critical counterweight.

If you're interested in pursuing this as a research angle yourself, the gap in the literature seems genuine: a systematic application of Popper's distinction between causal laws and historical trends to the Clements–Gleason debate, with attention to the teleological structure of climax theory, does not appear to have been written as a standalone philosophical argument.



